

Bank risk and returns: did prompt corrective action make a difference?

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Abstract

Purpose – The purpose of this paper is to assess the effects of prompt corrective action on bank risk and returns in an empirical framework.

Design/methodology/approach – The paper uses a difference-in-difference specification to analyse whether and how PCA affects bank risk and returns. As part of robustness, the analysis also uses a fixed effects specification with Driscoll–Kraay standard errors to account for serial correlation and cross-sectional dependence.

Findings – The findings reveal that banks under PCA framework contribute less to systemic risk and exhibit higher market valuation. These findings differ across recapitalised versus non-recapitalised banks and for banks with differing asset quality, capital and profitability. The overall price impact is a decline in lending rates and deposit costs.

Originality/value – To the best of the author's understanding, this is one of the early studies in the Indian context to carefully examine the linkage between PCA and bank behaviour.

Keywords Prompt corrective action, Systemic risk, TobinQ, State-owned banks, India, Tobin's Q

Paper type Research paper

1. Introduction

The health of the banking sector is of utmost importance for systemic stability, especially in economies where banks are the mainstay of financial intermediation. This assumes even greater relevance in emerging economies where state-owned banks dominate and are often used as conduits for implementation of government policies (Micco *et al.*, 2007; Cole, 2009; Baum and Caglayan, 2010; Bertay *et al.*, 2015; Morck *et al.*, 2019; Kumar, 2020; Mirzaei *et al.*, 2021). Repeated misuse of the state-owned banking sector for political expediencies coupled with inadequate risk assessment skills comprises on their asset quality. In turn, it weakens their capital position and lowers profitability. With inadequate fiscal resources, governments often “kick the can down the road” (Acharya *et al.*, 2021), leading to regulatory forbearance and weakening credit supply.

The Indian evidence is a case in point. Factors such as irrational exuberance in lending, malfeasance and lower economic growth worsened banks' asset quality over time (Rajan, 2018). As a reflection, gross non-performing loans of banks increased six-fold, from US\$21bn (1.3% of GDP) in 2011 to US\$121bn (4.4% of GDP) in 2020 (Reserve Bank of India, 2020). The problem was all the more acute for state-owned banks, which accounted for on average close to 85% of the banking system bad loans during this period. This has resulted



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in serious capital shortfall for Indian banks, an aspect that has been consistently highlighted by international organisations (IMF, 2014, 2017; Fitch, 2021).

As a long-term redress to improve the health and stability of banks, the Indian central bank introduced a Prompt Corrective Action (PCA) framework in 2017. The framework sought to incentivise banks to take timely corrective measures to restore their financial health (Reserve Bank of India, 2018). How far did the PCA achieve its intended objectives remains an open question.

To address this debate, we collate annual data on Indian state-owned banks during 2011–2020, to answer the following three questions. Firstly, did the PCA framework affect bank risk and relatedly, its returns? Secondly, did the impact differ across recapitalised versus non-recapitalised banks? And thirdly, what was the impact on bank lending and deposit rates? It was during this decade that this framework was implemented for banks which dictates the choice of this period.

Our empirical specification exploits the staggered implementation of the PCA across banks. We find that, on average, the PCA framework led to a decline in the systemic risk contribution of the bank, along with an improvement in their market valuation. The magnitude of the effect differed across banks recapitalised in a given year versus those which were not. These findings are valid after onboarding various robustness checks and indicate that PCA framework exerted a discernible impact on bank behaviour.

These are relevant questions for India for several reasons. Firstly, unlike the experiences of other countries, the banking sector still remains largely state-owned: the share of these banks in total banking assets averaged around 85% during 2011–2020. Although the ownership of these banks has been broad-based by allowing them to access equity capital, the government still retains majority ownership control. Reflecting these considerations, the levels of private credit are low as compared with other countries: the average levels of credit-to-GDP are around 50% as compared with 70% in Brazil and over 100% in Malaysia. This has occurred despite the country's gross domestic savings, at nearly 30%, being in line with peer countries. Thus, although the savings are sufficient, it is not allocated efficiently (Malpass, 2019). To reach the goal of a US\$5tn by 2025 as envisaged by the authorities, there is a need for stronger credit growth supported by diligent risk-taking. Secondly, robust economic growth had in the past led to lending over-optimism and in the process, hindered the development of adequate risk assessment skills (Loukoianova and Yang, 2018). However, as growth weakened, these projects faced bottlenecks and cost overruns, impeding firm cash flows and manifesting in burgeoning delinquent loan levels, threatening the fabric of financial stability (Rajan, 2017; IMF, 2018). As one of the possible responses, the Indian authorities initiated a PCA framework to undertake corrective measures in a timely manner. Thirdly, the government has also exhibited demonstrable commitment by introducing relevant changes in the Banking Regulation Act, 1949 (i.e. the Act governing banking companies). In particular, it has inserted two new Sections (namely, 35AA and 35AB) in the Act which empowers the Indian central bank to issue directions for resolution.

We make several useful contributions to the literature. Firstly, we assess the relevance of the PCA framework in the Indian context, a country with an overwhelming dominance of state-owned banks where the PCA framework has been recently enacted. In response to the burgeoning increase in loan delinquency over the past decade, the Indian central bank has undertaken a whole host of measures such as a repository on large credit to track correlated borrowers across banks in 2014, a framework for reviewing banks' asset quality in 2015 and the Insolvency and Bankruptcy Code for resolving corporate distress in 2016. At the same time, it has also been exhorting the government to infuse capital in state-owned banks. Notwithstanding these measures, bad banks of state-owned banks have continued to mount, posing challenges to financial stability. To complement these measures, the PCA has been

introduced. Several studies have reported that the PCA framework has been useful in lowering bank risk (Aggarwal and Jacques, 2001; Shimizu and Ly, 2017), others found this evidence to be less compelling (Ap Gwilym *et al.*, 2013). Eisenbeis and Kaufman (2008) suggested a structured intervention process to efficiently resolve bank failures. Valverde (2010) underscored the importance of coordination, both within country and across borders to arrest potential deterioration of banks' balance sheet and coming under the purview of PCA. As emphasised by Barth *et al.* (2016), PCA can tripwire during periods of crises and during such periods, standard regulatory measures provide much better signals of banking weaknesses (Gaul and Palvia, 2013). Our analysis shows that the PCA framework was useful in lowering bank risk and improving their returns.

Secondly, the study contributes to the literature that focuses on the behaviour of bank capital and relatedly, on their non-performing loans and profitability (Allen *et al.*, 2011; Chaibi and Fiti, 2015; Albertrazzi *et al.*, 2021). The regulatory hypothesis observes that regulators encourage banks to increase capital, commensurate with an increase in risk. Consistent with this hypothesis, studies show that risk and capital are positively related (Berger *et al.*, 1995; Jokipii and Milne, 2011). In contrast, the charter value argument contends that higher capital requirements lower bank charter values, impelling them to take higher risks (Hellmann *et al.*, 2000; Lindquist, 2004; Ding and Sickles, 2019). Although we focus on bank capital and risk, our focus is different in the sense that we demonstrate how PCA impacts banks with varying capital, profitability and asset quality metrics. The results indicate that the impact is most pronounced on banks with higher capital levels, whereas the impact on banks with weaker asset quality and low profitability are broadly similar.

Thirdly, the analysis augments the literature on bank recapitalisation (Giannetti and Simonov, 2013; Chodorow-Reich, 2014; Poczter, 2016; Berger and Roman, 2017; Berger *et al.*, 2019; Chopra *et al.*, 2021; Ghosh, 2022). As the PCA framework was operating in tandem with the recapitalisation process, it enables us to examine whether the impact of PCA differed across recapitalised versus non-recapitalised banks. The findings suggest that the impact of PCA was generally more effective for recapitalised banks, although in some instances, there was a discernible impact on non-recapitalised banks as well. The evidence concurs with recent research on corporate finance which show that, by facilitating access to credit for viable businesses at a lower cost, bankruptcy reforms improved the performance of distressed firms (Bose *et al.*, 2021).

The paper uses a measure of systemic risk that has gained acceptance in recent research (Acharya *et al.*, 2012; Brownlees and Engle, 2017). In essence, the measure captures the amount of capital that a bank requires during distress situations to maintain the stipulated capital ratio. For each bank, this relevant measure is normalised by its market capitalisation during the year (Berger *et al.*, 2022). Internationally, several studies have used this variable (Fuss and Ruf, 2021; Diez-Esteban *et al.*, 2022). Admittedly, this is one of the early studies to use this measure in the Indian context.

The rest of the article unfurls in the following fashion. In Section 2, we provide an overview of the relevant literature. Next, we highlight the key features of the PCA scheme. Section 4 describes the data and empirical strategy, followed by a discussion of the results including robustness and finally, we gather the concluding remarks.

2. Prior evidence

The theoretical evidence underlines the optimal trade-off between forbearance and corrective action. Using a dynamic framework, Kocherlakota and Shim (2007) find that PCA is preferred only when the likelihood of a decline in the collateral value is limited, otherwise forbearance is the preferred option. This evidence contrasts with Shim (2011) who concludes

that PCA is always optimal, *ex ante*. To stem possible contagion, [Morrison and White \(2013\)](#) argue that when the regulatory ability is low, forbearance might be the optimal strategy to support weak banks. As compared to this, [Qian et al. \(2019\)](#) observes that higher regulatory ability lowers the likelihood of bank risk-taking and in the process, dampens the need for forbearance.

Supervisory interventions to address bank weaknesses can either be under regular framework or under formal intervention regimes ([Svoronos, 2018](#)). In case of the former, supervisors initiate action based on their assessments regarding the health and stability of a bank. Although the combination of variables assessed varies across jurisdictions, without loss of generality, the variables typically considered include capital adequacy, the various facets of risks, governance, risk management procedures and profitability.

Formal intervention regimes are typically more stringent than regular supervisory interventions. These regimes specify the variables of supervisory concern and how their breach can trigger supervisory action. In addition, they also highlight the set of mandatory and discretionary actions the supervisor can enforce on the banks once these thresholds are breached. Salient among these are capital restoration plans, dividend curtailment and restricting the growth of risk weighted assets along with well-defined deadlines for completion and reporting of such actions.

The inspiration for formal intervention originated from the savings and loan crisis in the early 1990s. Given the spate of bank failures during this period, banking regulators in the USA adopted a structured early intervention and resolution process to address further systemic distress ([Benston and Kaufman, 1990](#)). Performance thresholds (based on capital) were used to categorise banks. A set of mandatory and discretionary actions was specified, with the intensity of the regulatory discretion being much stringent for banks significantly below the defined threshold. Since then, several countries both in Asia (e.g. India, Indonesia, Japan, Korea) and elsewhere (e.g. Chile, Mexico and Peru) have introduced such framework. A comprehensive and effective early intervention arrangement was adopted by the European Union in 2014, as part of the Bank Resolution and Recovery Directive (BRRD). This contrasts with the UK evidence where banks have to maintain target capital levels that are higher than the trigger capital level that can initiate supervisory action ([Ediz et al., 1998](#)). On the other hand, the framework for banks in Canada comprises of a four-tier qualitative framework, with each subsequent stage that coincides with a worsening of financial conditions involving a set of conditions for initiation of supervisory action ([Nieto and Wall, 2007](#)).

Academic interest in the PCA evolved since the early 1990s. In an early exercise, [Mingo \(1994\)](#) concluded that helping troubled banks through governance or ownership changes could facilitate capital infusion, if PCA was imposed on these banks. Subsequently, several studies studied the impact of potential breach of capital requirements for US banks. The broad conclusion is that prompt and structured equity infusion helps to augment banks capital and enable them to accommodate greater risk to extend credit ([Dahl and Spivery, 1995](#); [Jones and King, 1995](#); [Balla et al., 2015](#)), although greater proclivity towards risky choice regarding their basic business lines has provided detrimental ([DeYoung and Torna, 2013](#)). Focusing on federal deposit insurance corporation (FDIC)-insured commercial banks with minimum asset base of US\$100m during 1990–1997, [Aggarwal and Jacques \(2001\)](#) found that the PCA framework impelled banks to improve their capital base without offsetting credit risk increase, underscoring its efficacy in dealing with troubled institutions. Within a cross-national framework, [Chen et al. \(2021\)](#) find that banks subject to PCA tend to exhibit high systemic risk.

Subsequent analyses have emphasised other facets of PCA. Salient among these include the need for improved home-host supervisory coordination ([Valverde, 2010](#)) and more

involved measures to compute capital buffers comprising both stock (equity) and flow (loan loss provisions) components (Chernykh and Cole, 2015). Other suggestions include the incorporation of additional (non-capital) triggers such as those relating to asset quality or breach of large exposure (Svoronos, 2018; Cole and White, 2017) or even revamping prudential regulation to address regulatory forbearance during periods of correlated distress such as the financial crisis (Loveland, 2016). Garcia (2010) reviewed the reasons for failure of PCA during 2007–2009 and observed that supervisory forbearance of banks allowed them to remain solvent, although in most cases, banks regulatory capital was well below the supervisory requirements. Harris and Raviv (2014) suggest that an effective PCA framework requires a carrot-and-stick approach wherein banks are rewarded for reporting weaknesses while limitations are imposed on several of their activities. Kupiec (2016) provides a critique of the PCA framework in the US by observing that regulatory capital, which is a key constituent of the process, might not provide an accurate and timely reflection of the true financial health of a bank and therefore, might end up delaying the promptness of the corrective action. Ap Gwilym *et al.* (2013) show that although PCA effectively lowers default risk, its efficacy in lowering credit risk is much less compelling. Most recently, Kashyap *et al.* (2022) examined the relevance of the PCA framework and found that the mechanism effectively addresses borrower runs and safeguards financial stability.

Outside of the USA, studies on PCA have also been undertaken for other jurisdictions. Aglietta and Scialom (2010) argued that European banks' extensive cross-sector and cross-border outreach necessitates a holistic approach integrating prudential policy, market discipline and bank resolution. Likewise, Cetin (2011) argues that imposing prudent regulatory requirements on Turkish banks along with penalties can help foster a strong and safe banking system. Under the framework, the penalties can be of two types, institutional (e.g. prompt corrective actions, closures and even financial penalties) and personal (e.g. temporary prohibition from employment in the banking sector). Evidence from Korea examine the drivers that drive savings banks towards PCA and show that credit over-extension and poor regional economic performance are the key drivers (Sohn and Sim, 2014). In case of Japan, Shimizu and Ly (2017) observed that PCA intervention was useful and effective in stabilising systemic risk.

Few studies analytically assess the impact of PCA in the Indian context. Acharya (2018) provides graphical evidence which highlights that without the imposition of the PCA framework, Indian banks would have incurred much higher losses than is presently the case. Sumathy and Venkatachalam (2019) analysed the impact of PCA on banks and offered suggestions to lower the quantum of bad loans. Using information on the BRICS as reported in the cross-country bank regulation and supervision data developed by the World Bank, Gulati (2020) showed that the PCA regulations in India are comparable in terms of stringency to several other nations, although they fall short on other categories such as activity restrictions and insolvency powers. Bhusan *et al.* (2022) highlight the challenges involved in using the PCA under the current three-variable approach and offers alternate possibilities to ensure its seamless functioning. These studies are either documentary in nature or even if they are analytical, focuses on a small time window that limits their empirical relevance. We complement these studies by focusing on a larger time span, encompassing the entire gamut of state-owned banks, while taking into account other bank-specific characteristics that are relevant in influencing their behaviour.

3. Indian context

The Indian central bank had conceptualised a PCA framework in December 2002 on the lines of the FDIC's PCA framework. Unlike the FDIC framework, in the Indian context, it

also includes two additional parameters: asset quality and profitability. As [Acharya \(2018\)](#) has observed, given the low coverage for loan loss provisions, the level of bank capital, even if it is above the regulatory minimum, is not adequate enough to cover for even anticipated losses. To cover for potential capital write-offs that can occur in the future, these additional indicators – non-performing loans (to cover for asset quality weaknesses) and profitability (to cover for potential capital accretion) – have been incorporated. However, the framework was kept in abeyance. The Great Moderation of the early 2000 improved bank performance overall, relegating its importance. Subsequent developments such as the global financial crisis impacted the banking sector overall and therefore, regulators adopted a holistic strategy towards safeguarding systemic stability rather than focusing on individual banks.

In 2017, the Indian central bank issued the revised PCA framework and made it applicable for all banks operating in India, without exception. A major feature of the new framework was three sets of risk threshold for each variable (i.e. capital, NPL and RoA) to trigger mandatory and discretionary actions against the bank. Besides nurturing the banking system back to health through a set of mandatory and discretionary measures, the framework also provided the Indian central with the opportunity to interact more closely with bank management and sensitise them on the need to eschew riskier activities and conserve capital. The risk thresholds for each variable were explicitly outlined along with the set of actions under each of these thresholds.

To give some examples, when capital exceeds 8.375% but is below 10.875% (or alternately, the NPL levels exceed 6% but is less than 9%), mandatory actions such as dividend restrictions and capital infusion requirements kick in along with discretionary actions such as special supervisory monitoring meetings and targeted inspections at quarterly (or, other defined) frequency. There are also several strategy- and governance-related restrictions that become effective. Provided the bank becomes seriously undercapitalised (i.e. capital adequacy falls below 4.25%) or becomes excessively risk (i.e. loan delinquency ratio exceeds 12%), in addition to the measures outlined earlier, there are restrictions on branch expansion and management compensation (mandatory actions), in addition to credit- and market-risk related restrictions, including on their functioning and governance (discretionary actions).

In his deposition before the Parliamentary Committee on Finance in 2018, the Governor summarised the relevance of the PCA framework as follows:

It is important that public sector banks (PSBs) not simply be recapitalised but also be reformed operationally and financially to be rehabilitated to good health in a sustainable manner. The [PCA] framework aims at encouraging banks to eschew certain riskier activities and focus on conserving capital so that their balance sheets can become stronger. ([Patel, 2018](#)).

Table 1 provides a snapshot of the state-owned banks and the PCA impact. These banks on average accounted for 70% of total banking sector assets during the period 2011–2020. A minimum of one to a maximum of 11 banks have been under the PCA framework, with the share of affected equity being in the range of 3%–31%. In addition, these banks accounted for on average about a quarter of non-performing loans and close to 40% of the amount expended on their recapitalisation during this period.

As part of the empirical analysis, we focus on the period 2011–2020. We choose this period for three reasons. First and most importantly, it was during this period that the revised PCA framework was introduced. Secondly, the period also led to a rethinking of the governance and business structures in state-owned banks in the wake of several instances of financial misconduct and lending exuberance ([Reserve Bank of India, 2014](#); [Kumar *et al.*, 2016](#); [Rajan, 2018](#)) and in that context, how better to improve the efficacy of bank resolution

Table 1.
Summary statistics
of state-owned
banks, by year

Year	N. banks	Asset share (in total banking sector, %)	N. banks under PCA	RoA (%)*	Share (%) in state-owned banks overall		
					Gross NPL	Capital	Recapitalisation
2011	21	74	0	0	0	0	0
2012	21	73	0	0	0	0	0
2013	21	73	0	0	0	0	0
2014	21	73	0	0	0	0	0
2015	21	72	0	0	0	0	0
2016	21	70	0	0	0	0	0
2017	21	69	1	−0.1	3.4	2.6	1.8
2018	21	66	10	−1.7	38.0	26.2	55.9
2019	21	62	11	−2.5	39.7	31.0	69.3
2020	21	60	5	−2.2	12.0	9.8	25.3

Note: *Average (if, more than one bank)
Source: Author’s calculations

framework. And finally, the period coincides with the post global financial crisis, providing a database that is less susceptible to exogenous influences.

4. Database and research design

We use three sets of variables at the bank level. The first set comprises bank financials. We observe data on balance sheet and financials for state-owned banks covering the period 2011–2020 from the *Statistical Tables Relating to Banks*. We complement this with bank-level information on prudential figures such as capital adequacy and non-performing loan ratios from the *Report on Trend and Progress of Banking*. The second set comprises information on banks subject to PCA, which we obtain from the various documents of the Indian central bank (Reserve Bank of India, 2018, 2020). Thirdly, the *Prowess* database (akin to US *Compustat*) provides information on banks’ annual closing share price and number of shares outstanding. This helps to calculate the market value of equity and relatedly, TobinQ.

We make three adjustments to the data. Firstly, data on IDBI Bank is considered till 2019 consequent upon the transfer of its majority ownership away from the government from 2020, thereby changing its ownership to a private bank. Secondly, we do not include the associate banks of the State Bank of India, because all of them were merged with the State Bank of India at various time points during this period [1]. Finally, we include dummies to control for mergers that occurred during this period. Consequent upon these adjustments, we have a maximum of 207 bank-years. To account for potential outliers, all bank-specific variables are winsorised at the top and bottom 1 % levels.

A key variable in our framework is systemic risk. This is calculated using the following equation:

$$SRISK_{b,t} = k Debt_{b,t} - (1 - k) (1 - LRMES_{b,t}) Equity_{b,t} \tag{1}$$

where k is the book equity to asset ratio (set equal to 8%, following Brownlees and Engle, 2017), $LRMES$ is the long-run marginal expected shortfall, that defines the bank’s expected loss under a crisis scenario [2]. $SRISK$ therefore shows the firm’s contribution when there is systemic distress. Intuitively, during periods of idiosyncratic distress (i.e. a bank’s equity

capital falls short of its debt- or debt-like liabilities), the systemic economic consequences are limited, because healthier banks can compensate for the intermediation void of the failed bank. On the other hand, during periods of systemic distress, the overall capitalisation declines, leading to a breakdown in aggregate intermediation services, because other financial entities are not equipped to address this void. Such systemic disruptions have severe ramifications for the broader economy including credit crunch and distress sale of assets. We obtain this measure for each bank-year from *V-Lab* (the volatility laboratory at New York University that provides real-time measurement and forecasting of financial volatility and correlations across a wide spectrum of assets) [3]. Akin to [Berger et al. \(2022\)](#), we scale this variable by the respective bank's market capitalisation during the year. This measure, termed *NRISK* shows the proportional increase in capital that a bank would need during distress periods.

[Table 2](#) shows the variable definitions and summary statistics. We use two market-based measures of risk and returns: *NRISK* and *TobinQ* ([Weir et al., 2002](#); [Jones et al., 2011](#); [Berger et al., 2020](#); [Nawaz et al., 2020](#)). Among others, we also include the bank-specific lending and deposit rates. The average level of *SRISK* is 1.9, so that during periods of distress, a bank on average contributes 2% to overall systemic risk. Likewise, banks exhibit higher market valuation (as proxied by their Tobin's *Q*). Besides their forward-looking nature that helps internalise the expectations of shareholders regarding future behaviour, they are less susceptible to accounting conventions and other frictions (e.g. tax laws) that permeate accounting measures ([Wernerfelt and Montgomery, 1988](#); [Demsetz and Villalonga, 2001](#)).

Variable	Definition	Obs.	Mean (SD)	p. 25 (p. 75)
<i>Dependent</i>				
Systemic risk (NRISK)	Systemic risk measure developed by Acharya et al. (2012) and refined by Brownlees and Engle (2017) normalised by the bank's market capitalisation	207	1.874 (1.402)	0.830 (2.650)
TobinQ	(Market value of equity + book value of debt) / Total asset	207	0.101 (0.045)	0.071 (0.118)
Lending Rate (LendRT)	Interest earned on advances / Total advances	207	0.090 (0.010)	0.080 (0.099)
Deposit rate (DepRT)	Interest expended on deposit / (Total deposits - demand deposits)	207	0.062 (0.009)	0.054 (0.071)
<i>Independent</i>				
PCA	Dummy = 1, for the years a bank is under PCA framework, else zero	209	0.129 (0.336)	0 (0)
Asset (LTA)	Ln (Asset)	207	17.320 (1.033)	16.510 (17.900)
Capital adequacy ratio (CRAR)	Capital / Risk weighted asset	207	0.117 (0.011)	0.110 (0.126)
Cost-income ratio (CIR)	Operating expenses / (Total income - interest expense)	207	0.503 (0.090)	0.440 (0.560)
Fee income (FEE)	Non-interest income / Total asset (%)	207	0.874 (0.248)	0.699 (0.98)
Recapitalisation (RECAP)	Amount recapitalised / Asset (%)	207	0.541 (0.762)	0 (0.732)
Merger	Dummy = 1 for the acquirer bank in the year of merger	209	0.005 (0.069)	0 (0)

Source: Author's calculations

Table 2.
Variable definition
and summary
statistics

Among the independent variables, the natural logarithm of total (real) asset was 17.3 on average, which translates into INR 2,500bn. Banks are well-capitalised, with high non-performing loans, on average. Their efficiency levels, as measured by cost-to-income are high. The average recapitalisation has been 0.5% of total asset.

The key independent variable is PCA. As our earlier table 2 had shown, not all banks were subject to PCA during the period. This is reflected in the high standard deviation on this variable.

The correlation matrix of PCA with the independent variables in Table 3 shows a negative and statistically significant association with risk and a positive relationship with returns. This would suggest that PCA dampens both risk while improving returns, although the magnitude of the correlation is less than 10%. However, these raw correlations do not account for other unobservables and as a result, we use an empirical specification to tease out the relationship. However, as we do not have recourse to any natural experiment, the analysis at best highlights the evolution of bank risk and returns in response to the PCA.

To investigate the impact of PCA on bank behaviour, we estimate a specification of the following form:

$$y_{bt} = \nu_b + \eta_t + \alpha PCA_{bt} + \beta X_{bt} + \varepsilon_{bt} \tag{2}$$

where y measures bank risk or returns or a price variable (e.g. lending or deposit rate, in subsequent analysis), ν captures bank- fixed effects, η denotes year fixed effects and ε is an idiosyncratic error.

PCA is a dummy that equals one for the years a bank is under PCA, else zero. The coefficient we focus on is α : it shows the response of banks subject to PCA versus those which are not, on the outcome variable. A statistically significant value on α would signify that PCA exerts a perceptible impact on the outcome. Throughout, standard errors are double-clustered by bank and year.

The vector X represents bank-specific controls, such as size, funding structure, income profile and cost efficiency (Berger *et al.*, 2005). Large banks have the ability to take higher risks, supportive of a positive relationship (Laeven *et al.*, 2014; Bhagat *et al.*, 2015). Banks with lower equity capital are likely to have less stable funding and therefore, more prone to risk-taking, consistent with the charter value argument (Hellmann *et al.*, 2000; Laeven *et al.*, 2014). Higher income diversity permits banks to tolerate lower risk, because of multiple sources of revenue. Finally, banks with higher cost-to-income are more susceptible to risk-taking to compensate for operating inefficiencies.

We proceed as follows. Firstly, we estimate the baseline regression for the various outcome, first without the control variables and incorporating them subsequently. Next, we focus on recapitalised versus non-recapitalised banks. The extant evidence reveals that

Table 3.
Correlation matrix of
key variables

	PCA	NRISK	TobinQ	LendRT	DepRT
PCA					
NRISK	−0.042**				
TobinQ	0.078**	−0.374***			
LendRT	−0.196***	0.127*	−0.154**		
DepRT	−0.265***	0.225***	−0.079	0.848***	

Note: *** ** and *indicate statistical significance at 1, 5 and 10% levels, respectively
Source: Author’s calculations

recapitalisations are associated with an increase in lending (Giannetti and Simonov, 2013). That being the case, this should improve bank valuation and relatedly, lower risk-taking. Thereafter, we undertake additional robustness checks and finally, look at lending and deposit rates.

5. Discussion of the results

5.1 Main findings

The main findings are presented in Table 4. As discussed earlier, in all cases, we first run the regression without the controls and subsequently include the control variables. Column 1 shows that the coefficient on PCA is negative and statistically significant with a point estimate of -0.62 . Inclusive of controls (Column 2), the magnitude of the coefficient remains statistically significant with a point estimate of -0.53 . Therefore, the contribution to systemic risk arising from the distress of a particular bank declines by 0.5% on average when it comes under the ambit of PCA. The findings resonate prior evidence for Japan where the contribution to systemic risk (measured by CoVAR) emanating to idiosyncratic distress was observed to decline when a bank was brought under PCA (Shimizu and Ly, 2017). Earlier, Ediz *et al.* (1998) had reported that the UK banks also readjust their capital levels when their capital levels decline towards trigger levels (i.e. bank-specific minimum capital requirement).

The evidence presented also resonates with evidence from the broader literature. For example, focusing on bank failures during 2007–2012 and provide evidence that although distressed banks had high levels of non-performing loans and inadequate loan loss reserves, their regulatory capital levels were sufficient to address potential exigencies. They conclude that regulatory forbearance allowed banks to bypass the disciplinary measures that are required for under PCA. Evidence from the USA finds that banks overstated the value of distressed assets to bolster their bottom lines and provide suggestive evidence that regulatory forbearance could be an important driver (Huizinga and Laeven, 2012).

Columns (3)–(4) focus on returns. Our preferred specification is Column (4) which includes bank-specific variables including the macroeconomic environment. We find that banks under PCA exhibit higher market valuation. This could occur because the market perceives that the regulatory intervention would nurture these banks to sound health, which

	NRISK		TobinQ	
	(1)	(2)	(3)	(4)
PCA	-0.623^{***} (0.261)	-0.530^{**} (0.261)	0.0004 (0.005)	0.010^{**} (0.005)
LTA		0.046^{***} (0.016)		-0.0008 (0.003)
CRAR		-0.857 (0.679)		0.296^{*} (0.163)
FEE		-1.070^{**} (0.499)		-0.016^{*} (0.009)
CIR		0.162^{***} (0.042)		-0.050^{**} (0.025)
Merger	NO	YES	NO	YES
Bank FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N.Obs	207	207	207	207
Adj. R-sq.	0.6125	0.6166	0.8034	0.8179

Notes: Robust standard errors (clustered by bank-year) in brackets. ***, ** and * indicate statistical significance at 1, 5 and 10% levels, respectively

Source: Author's calculations

Table 4.
PCA and Bank
behaviour – main
results

is viewed as credit positive. In terms of magnitude, banks subject to PCA have 1% higher market valuation, on average.

Collectively, these findings add to the evidence which shows that PCA exerts a discernible and salutary impact on bank behaviour.

The sign of the control variables indicates the following. Firstly, bigger banks have higher levels of risk, consistent with prior evidence (Laeven *et al.*, 2014). Economically, a one standard deviation increase in bank asset increase risk by roughly 17%. Well-capitalised banks assume lower risks, consistent with the moral hazard argument that higher capitalisation makes banks risk-averse, because the downside risks far outweigh the upside (Jacques and Nigro, 1997; Tan and Floros, 2013; Ding and Sickles, 2019). On the other hand, better capitalised banks have higher market valuation, consistent with the argument that such capital improves their lending capability and offers opportunities for growth. Fee income exerts a negative and statistically significant impact on bank returns. This could be attributed to the fact that such incomes are usually more volatile and can therefore lead to higher profit volatility (DeYoung and Roland, 2001). Higher levels of inefficiency increases risk and lowers market valuation of banks, consistent with the skimping hypothesis that banks compromise on short-term operating expense to achieve efficiency gains, which eventually results in higher credit risk and relatedly, provisioning and thereby lower capital, which raises systemic risk (Berger and DeYoung, 1997; Fiordelisi *et al.*, 2013). Higher GDP growth lowers risk and raises bank valuation, as expected.

6. Robustness checks

6.1 Recapitalised vs non-recapitalised banks

The PCA process was operating in tandem with the bank recapitalisation program. Indeed, as Table 1 shows, during 2017–2020, PCA banks accounted on average for around 40% of the recapitalisation support provided to state-owned banks. It therefore remains to be explored whether the risk return profile of recapitalised banks under the PCA framework differed from those that were not recapitalised. To investigate this, we re-estimate the previous specification, separately for these two categories. The results are presented in Table 5.

We find that non-recapitalised banks displayed a much higher contribution towards lowering systemic risk as compared with recapitalised banks. To see this, note that the coefficient on PCA in column (1) is -0.43 and the corresponding coefficient in column (3) is -2.22 . In other words, the contribution to systemic risk arising from the distress of a

Table 5.
PCA and Bank
behaviour –
recapitalised vs non-
recapitalised banks

	Recapitalised banks		Non-recapitalised banks	
	NRISK (1)	TobinQ (2)	NRISK (3)	TobinQ (4)
PCA	-0.431^{***} (0.127)	-0.002 (0.007)	-2.221^{**} (0.914)	0.039^{***} (0.013)
Controls	YES	YES	YES	YES
Bank FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N.Obs	147	147	55	55
Adj. R-sq.	0.6401	0.8433	0.6212	0.8010
Notes: Robust standard errors (clustered by bank-year) in brackets. ***, **and * indicate statistical significance at 1, 5 and 10% levels, respectively				
Source: Author's calculations				

non-recapitalised bank under PCA is five times larger in magnitude as compared with a recapitalised bank under PCA. Economically, non-recapitalised banks have relatively higher capital base; for them, the challenge is more from the other two parameters such as asset quality and/or profitability. Therefore, when these banks come under PCA, their contribution towards reduction in systemic risk (which, among others, is a function of their capital) is much higher. Non-recapitalised banks are also observed to display higher growth opportunities, consistent with the argument that their stronger capital base facilitates higher lending and profitability, which is likely to be manifest in higher growth.

Collectively, these findings suggest that PCA exerts a perceptible impact on the risk and return matrix of recapitalised versus non-recapitalised banks.

6.2 Checking for reverse causality

The cross-sectional heterogeneity partially addresses potential concerns that it is PCA that influences bank behaviour and not the other way around. To confirm this further, we include dynamism into the framework. To implement this, we replace the PCA variable PRE1 and PRE2: these are the one- and two-year lagged values of PCA. The contemporaneous impact is captured by PCA. The remaining variables are similar to earlier.

From an economic standpoint, if the coefficient on PRE is statistically significant, this would suggest that there is an impact on banks before the introduction of PCA and suggest the presence of reverse causation (Acharya *et al.*, 2013). Table 6 shows that the coefficient on PRE1 is not statistically significant and as a result, the possibility of reverse causation can be ruled out. Besides, we find that the coefficient on PCA is statistically and economically significant in both cases and in line with the estimates obtained in the baseline model. This supports the fact that it is changes in PCA lead to changes in bank behaviour.

6.3 Serial correlation and cross-sectional dependence

There is a possibility of sequential correlations and cross-sectional dependence. We use the Pesaran (2004) test to determine the presence of cross-sectional dependence in the data. The null hypothesis is no cross-sectional dependence and the test statistic converges to a normal distribution as the cross-section goes to infinity and the time series increases. The Pesaran test statistic was statistically significant at the 0.05 level, indicating cross-sectional dependence.

	NRISK (1)	TobinQ (2)
PRE2	Omitted	Omitted
PRE1	−0.093 (0.359)	−0.008 (0.008)
PCA	−0.552** (0.251)	0.015** (0.006)
Controls	YES	YES
GDPGR	YES	YES
Bank FE	YES	YES
N.Obs	207	207
Adj. R-sq.	0.6144	0.8179

Notes: Robust standard errors (clustered by bank-year) in brackets. ***, ** and * indicate statistical significance at 1, 5 and 10% levels, respectively

Source: Author's calculations

Table 6.
PCA and Bank
behaviour – reverse
causality

To address this concern, we use Driscoll–Kraay standard errors, which are robust against autocorrelation, heteroscedasticity and cross-sectional dependence. The lag length is chosen based on [Hoechle \(2007\)](#) and is based on the first step of the [Newey and West's \(1994\)](#) plug-in procedure [4].

Results are set out in [Table 7](#). We find that there is a significant decline in risk for both recapitalised and non-recapitalised banks; in case of the latter, the magnitude is higher (in absolute terms), consistent with the baseline. Interestingly, the impact persists for multiple periods, suggesting that there is a longer-term impact of PCA on recapitalised banks. The evidence concurs with the USA wherein the findings suggest lagged effects of PCA on real outcomes ([Danisewicz et al., 2018](#)). In case of Tobin's Q, the impact is manifest only for non-recapitalised banks and there is both an immediate as well as a long-term positive impact of PCA.

6.4 Impact on returns and cost

Finally, we examine the impact of PCA on lending and deposit rates. Intuitively, there could be a “stigma effect” associated with banks under PCA. That being the case, this could influence their lending and deposit rates. Evidence of such an effect was evidenced for the US banks during the global financial crisis: banks were sceptical of accessing the Fed discount window because if it became known, it might be interpreted as a signal of weakness ([Armentier et al., 2015](#)). This impact would also likely differ for recapitalised and non-recapitalised banks, given the possibility of some kind of reputation risk associated with the former ([Fiordelisi et al., 2013](#); [Barakat et al., 2018](#); [Brei et al., 2020](#); [Adeabah et al., 2022](#)). Taking these considerations on board, we estimate the baseline specification for all banks and separately by their recapitalisation status. [Table 8](#) shows the estimates.

In columns (1) and (2), we find that banks under PCA lowered their lending rates by 6 basis points (bps) and deposit rates by a slightly higher 8 bps. Across categories, recapitalised banks lowered their rates *pro rate* by 7 basis points, presumably to avoid loss of market share to cover for reputational risk. As compared to this, the better capital position of non-recapitalised banks allowed them to lower their deposit rates and lending rates by a much higher magnitude than overall in expectations that can exit the PCA framework much quicker and garner market shares to fill the void in lending.

	Recapitalised banks		Non-recapitalised banks	
	NRISK (1)	TobinQ (2)	NRISK (3)	TobinQ (4)
PCA(t−1)	−0.219** (0.107)	−0.002 (0.007)	−1.056* (0.554)	0.048*** (0.021)
PCA (t−2)	−0.039*** (0.012)	0.006 (0.009)	−0.176** (0.069)	0.027** (0.013)
Controls	YES	YES	YES	YES
Bank FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
N.Obs	147	147	55	55
Adj. R-sq.	0.7219	0.5836	0.4897	0.6254

Table 7.
PCA and Bank
behaviour –
robustness

Notes: Driscoll–Kraay standard errors are within brackets. ***, ** and * indicate statistical significance at 1, 5 and 10% levels, respectively
Source: Author's calculations

7. Concluding remarks

To limit the potential adverse consequences arising from bank failures, authorities often resort to structured interventions. The underlying objectives are to nurture the system back to profitability in a time-bound manner and address systemic risks. However, whether and to what extent are these two objectives adequately met is not clear, *a priori*.

To inform this debate, we analyse the impact of PCA on risk and returns of Indian state-owned banks covering the period 2011–2020. The staggered nature of the intervention provides significant variation, both in the time-series and the cross-section of banks. The evidence supports the fact that PCA lowers systemic risk and improves bank value. These findings differ across recapitalised versus non-recapitalised banks and across banks with varying strengths of their capital and loan book. The net effect of these developments is manifest in a decline in lending rates and deposit costs.

The aforesaid analysis provides useful insights from a policy standpoint. First of all, safeguarding depositor interests is an important ingredient of a holistic banking supervision architecture. This assumes greater relevance in the current context because although they account for over 80% of the core funds for banks. In this context, corrective action needs to be undertaken to address financial instability and limit potential contagion. Viewed from this perspective, the analysis highlights that the PCA framework has been effective in limiting systemic weaknesses. Secondly, the evidence suggests that PCA can also have a signalling effect. The removal of banks from the PCA framework provides confidence to stakeholders regarding the health and stability of the banks. In particular, banks which are not recapitalised and able to successfully navigate the PCA framework emerge much stronger in terms of their internal controls, which helps to raise overall systemic stability. Thirdly, a negative fallout of banks being subject to PCA is the stigma effect, wherein they become subject to depositor discipline by paying higher costs. They have to lower their lending costs as well, to regain market shares. Such a negative fallout could end up weakening these banks, as it erodes their margin and longer-term viability. How best to tailor the PCA process to address this challenge needs to be explored further.

Several additional questions can be addressed in this regard. How did the lending behaviour of banks evolve in the post-PCA phase? Did banks improve their extensive margin (i.e. lending to new customers) or intensive margin (i.e. higher lending to existing borrowers, perhaps by adjusting their terms of credit)? Were there any changes in the intensive margin (i.e. terms of credit) as a result of the recapitalisation? Did such PCA affect firm–bank lending relationships? Such deeper analysis would necessitate more granular data, which can help to better inform the policy debate, going forward.

	All banks		Recapitalised banks		Non recapitalised banks	
	LendRT (1)	DepRT (2)	LendRT (3)	DepRT (4)	LendRT (5)	DepRT (6)
PCA	−0.006*** (0.002)	−0.008*** (0.001)	−0.007*** (0.002)	−0.007*** (0.001)	−0.014*** (0.003)	−0.016*** (0.003)
Controls	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Bank FE	YES	YES	YES	YES	YES	YES
N.Obs	207	207	147	147	60	60
Adj. R-sq.	0.5058	0.5304	0.5050	0.4980	0.3618	0.5849

Notes: Robust standard errors (clustered by bank-year) in brackets. ***, ** and * indicate statistical significance at 1, 5 and 10% levels, respectively

Source: Author's calculations

Table 8.
PCA and Bank
behaviour – impact
on deposit and
lending rates

Notes

1. State Bank of India is the largest bank in India in terms of financials. According to *The Banker*, in 2020, it was ranked 56th globally with \$38bn of tier-I capital, \$660bn of assets and serving over 450 million customers.
2. $LRMES = 1 - \exp(\log(-18 * MES))$, where $MES = \frac{1}{N_{days}} \sum_{\tau=1}^{\tau} R_{b,\tau}$ and R is the daily returns of the bank.
3. This statement is taken from V-Lab website. Available at [<http://vlab.stern.nyu.edu/about>].
4. The lag length is defined as $m(T) = \text{floor}[4 (T/100)^{(2/9)}]$, where T denotes the time period. In our case, it works out to be 2.49 (≈ 2).

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